DIGITAL HEALTH | WOMEN'S WELLNESS

CycleTwin

A Digital Twin of menstrual and overall wellness: a living virtual model of her body that learns from daily inputs and predicts what happens next.

Warm. Private. Empowering.



Sample Cycle Ring (illustrative)

Team: purvawinwin | College Name:Army institute of Technology,Pune

The Problem

Menstrual health is under-tracked and poorly understood

Fixed predictions

Most apps assume a 28-day cycle, which fails for irregular cycles.

Scattered data

Sleep, food, stress, mood and symptoms are tracked apart, so patterns stay hidden.

Late awareness

Signs of PCOS, thyroid issues or anemia get missed when girls feel shy or uninformed.

Weak doctor visits

Doctors get memory-based histories instead of clear data.

Trust and access

Tools feel clinical, lack privacy, and rarely support regional languages.

CycleTwin closes these gaps with one private, predictive twin.

Our Solution

From a 60-second daily log to a personal predictive twin

1

Log

Tap-based daily log: flow, pain, mood, sleep, food, exercise, symptoms.

2

Model

The twin updates the Cycle Ring, Wellness Score and Twin Health Age.

3

Predict

Adaptive predictions and gentle alerts: late period, PMS, low-iron risk.

4

Simulate

What-if Simulator shows how sleep, water, exercise or stress could change her day.

5

Act

Phase-synced meals and workouts, and a doctor-ready summary with her consent.

Private by design: consent-based sharing, and individual data is never sold.

Healthcare Use Case

Who it helps and how it is used

Who it helps

- Teens and young women
- Gynecologists and nutritionists
- Schools, colleges and NGOs (anonymous data)

Impact

- Earlier awareness of irregular patterns
- Better conversations with doctors
- Indian-friendly nutrition and plans
- English, Hindi and regional languages

Example scenario

1. Onboards: age, cycle, diet, activity
2. Logs daily in under a minute
3. Twin predicts the next period from her last 3-6 cycles
4. Low-iron alert; she tries an iron-rich plan in the simulator
5. Pattern flags say see a doctor; she shares a read-only summary

Key Features

18 connected modules in one app

Cycle Ring

Animated day and phase view

Digital Twin

Clickable body zones, hormone chart

Simulator

What-if energy, pain, mood

Insights

Trends, heatmap, radar, PDF

Wellness Plans

Phase-synced Indian meals, workouts

Ask Twin

AI chat, voice, multilingual

Mind Care

Breathing, meditation, journal

Risk Screener

PCOS, thyroid, anemia signals

Doctor Mode

One-page summary, consent links

Care and Learn

Helplines, red flags, myth vs fact

Hygiene Log

Products, reminders, cost

Privacy

PIN lock, incognito, delete data

Technical Stack

Modern, mobile-first and offline-friendly

Layer	Technology
Frontend	React, TypeScript, Tailwind CSS, React Router
Charts	Recharts: line, radar, heatmap, bar, donut
Backend / Auth	Supabase (auth + storage), localStorage fallback
App type	Progressive Web App, mobile-first, light and dark mode
AI	LLM-powered Ask Twin [add provider / model]
Tools	Lovable, VS Code, GitHub

18
modules

6
wellness axes

4
hormones estimated

60s
daily log

AI / ML and Logic

Transparent, explainable methods plus a conversational layer

Adaptive prediction

Rolling average of her last 3-6 cycles, not a fixed 28 days.

Wellness Score

Weighted mix of sleep, nutrition, activity, stress, regularity, hydration. Twin Health Age comes from it.

Hormone estimates

Estrogen, progesterone, LH and FSH curves from a cycle phase model. Illustrative, not lab-measured.

What-if Simulator

Recomputes predicted energy, pain, mood and score against her baseline.

Risk Screener

Rule-based low / medium / high pattern indicators. Always: please consult a doctor.

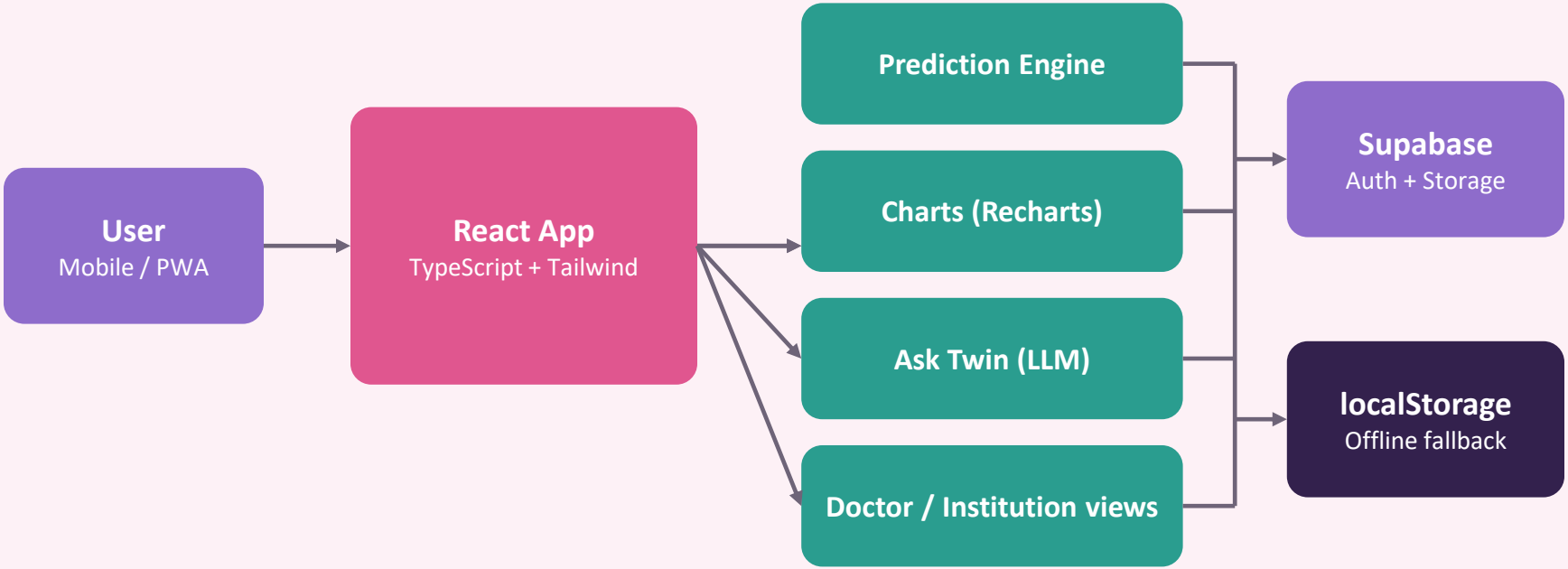
Ask Twin

LLM chat using her logged data, with voice input and multilingual replies.

Limitation: rule-based and statistical logic, not a clinically validated model. Outputs are educational estimates.

Architecture

How the pieces connect



Doctor and Institution views use consent-based sharing and anonymous aggregates only.

Business Model

Free for every girl; sustainable through premium and partners

Free

Always free

Tracking, daily log, predictions, calendar and education.

Premium

₹149/month or ₹999/year

7-day trial. Ask Twin, simulator, full reports + PDF, meal and workout plans, Doctor Mode, risk screener.

Institution

From ₹49/student/year

Anonymous aggregate analytics, bulk invites, licence management, impact report.

Other revenue streams

- **Doctor consult commission (20%)**
- **Labeled affiliate shop and sponsorships**
- **Grants and CSR**
- **Anonymized, consent-based research insights**

Privacy and Trust

Built around her consent and control

Her data, her call

Never sold. No ads in health logs or doctor summaries. Aligned with India's DPDP Act.

Lock and hide

PIN / biometric lock and Incognito mode that hides the app name.

Consent first

Per-use consent toggles and read-only doctor links with granular scope.

Your data, your call

Data export and one-tap delete.

DPDP-aligned

Privacy page aligned with India's DPDP Act.

Accessible

Large text, color-blind-safe charts, offline support.

Safe by design

Never a diagnosis. Red-flag checker and helplines 112, 181, 14416.

Roadmap

Planned path forward

1

Phase 1

Core tracker, twin, predictions and education

2

Phase 2

Pilot with schools and colleges; regional languages

3

Phase 3

Doctor marketplace and wearable integration

4

Phase 4

Clinical validation with doctors; wider rollout

The roadmap shows intended direction; timelines to be confirmed by the team.

Conclusion

CycleTwin is a smart digital health platform that helps girls track their menstrual cycles, understand their wellness patterns, and receive personalized predictions. With a focus on privacy, awareness, and better health decisions, it empowers girls to take control of their well-being